123

Simulating Tumor-Induced Angiogenesis in Physicell

Divyansh Rana[1], Ronak Monga[2] and Leena Sohail[3]

[1]Dept. of Computer Science, Indiana University, Bloomington, IN, USA [2]Dept. of Data Science, Indiana University, Bloomington, IN, USA [3]Intelligent Systems Engineering, Indiana University, Bloomington, IN, USA

*Corresponding Author - melovele@iu.edu

Introduction

Tumor-Induced Angiogenesis and PhysiCell Simulation

- Tumors require blood vessels to grow and metastasize
- Angiogenesis supplies oxygen and nutrients
- Removes waste from proliferating tumor cells
- Enables invasion into surrounding tissues and vasculature
- We use PhysiCell to simulate angiogenesis in 2D environments
- Captures complex, multicellular interactions in tissue microenvironments
- PhysiCell simulates tumor-driven vascular growth
- Tumor cells release VEGF in response to hypoxia
- VEGF diffuses through the tissue
- Endothelial cells detect VEGF gradients and initiate sprouting
- Sprouts extend toward hypoxic regions, forming vascular networks
- **Key simulation features:**
  - Tumor-endothelial interactions in a 2D matrix
  - Diffusion-reaction modeling of VEGF
  - Dynamic blood vessel formation
  - Environment continuously reshaped by cell behavior
- **Simulation goal:**
  - Understand how tumors exploit angiogenesis
  - Analyze vascular growth patterns
  - Explore therapeutic strategies to disrupt this process

Objective

We aim to model vascularization and angiogenesis by simulating interactions between tumor cells, endothelial cells, and VEGF. This allows us to study how tumors trigger new blood vessel growth under hypoxic conditions.

Methods

Simulation Setup

- Defined 4 main cell types: healthy tumor, hypoxic tumor, endothelial, static vessel
- Adjusted secretion rates for VEGF (tumor) and oxygen (vessel) to match literature
- Tuned cell behaviors for each type: growth, movement, response to VEGF and oxygen
- Added cell-cell adhesion to endothelial and vessel cells

Goal: promote realistic vessel network formation

- Blood vessels stick to nearby vessels to mimic branching behavior
- Modified motility settings to control movement speed of each cell type
- Simulated in 2D with periodic boundary conditions

Initial Conditions

- Healthy tumor placed in the center of the environment
- Initialized with a limited bolus of oxygen to sustain early tumor
- Initial vessels generated with a random walk python script
- Endothelial cells seeded at the tips of vessel branches
- VEGF gradients allowed to emerge dynamically
- Simulation run time: 20,000 minutes (~14 days biological time)
- Time steps adjusted to slow simulation and improve visual observation

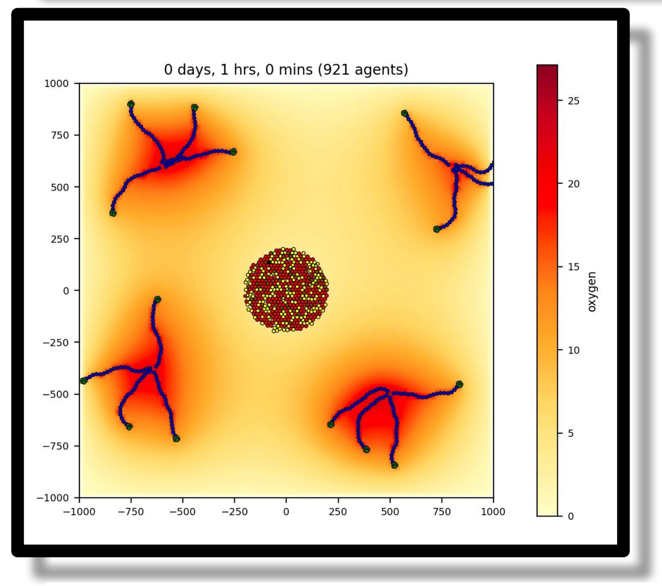
Endothelial Behavior Testing

- Varied endothelial cycle entry rate with VEGF to control proliferation
  - Reproduction:
    - 80% to leave a static blood vessel
    - 20% to create a new branching endothelial cell
- Compared two strategies:
  - More vessel cells = faster sprouting
  - More endothelial cells = stronger VEGF response
- Observed how VEGF gradients influenced growth direction and density
  - Element of randomness to endothelial motility for more of a "branching" effect
- Tracked number of sprouts and vessel connections per simulation

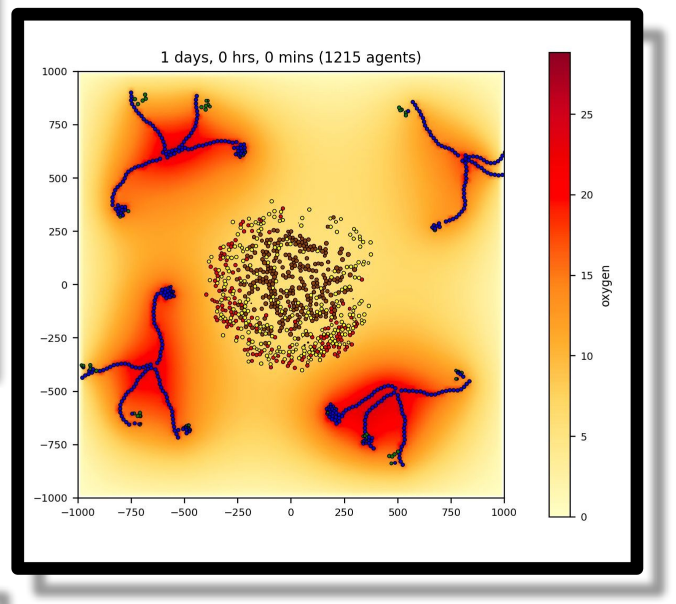
Evaluation

- Observed vascular coverage over tumor area
- Counted total number of vessel structures at end of simulation
- Visually inspected network formation quality (connectivity and spread)
- Discussed limitations and possible future modifications

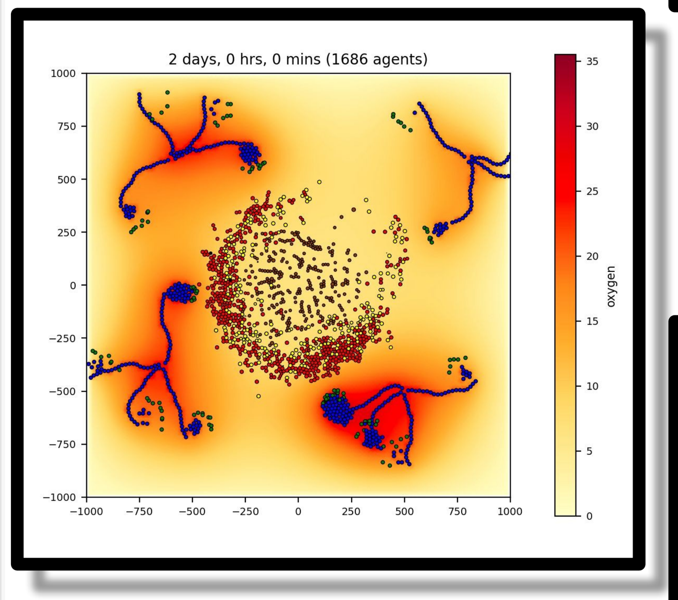
Results



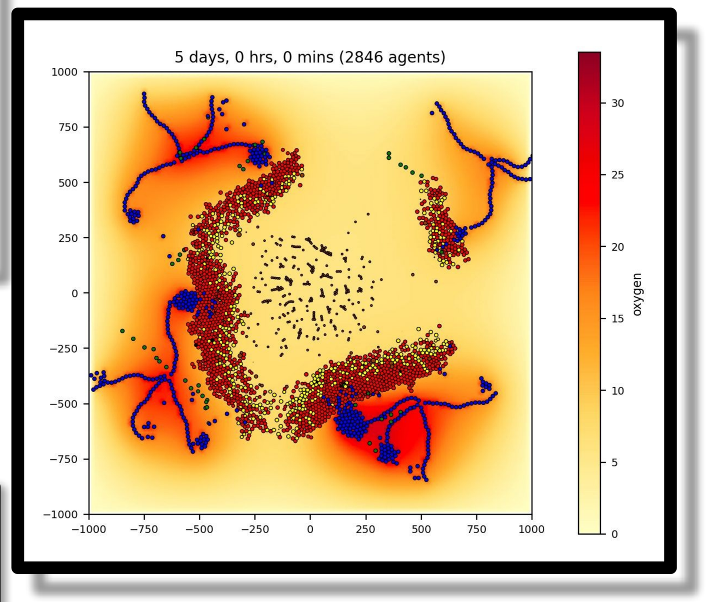
1 Hour: Tumor initialized at center with limited oxygen supply to mimic a realistic starting point



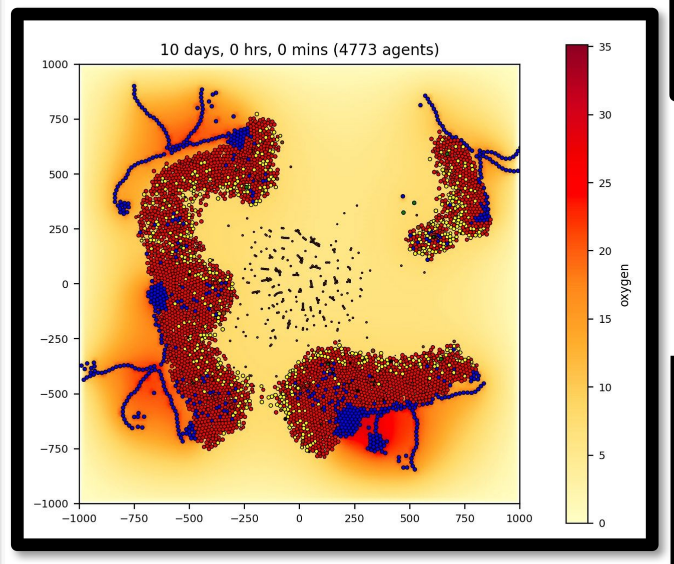
1 Days: Tumor cells in the core become hypoxic due to limited oxygen diffusion, and begins migrating outward



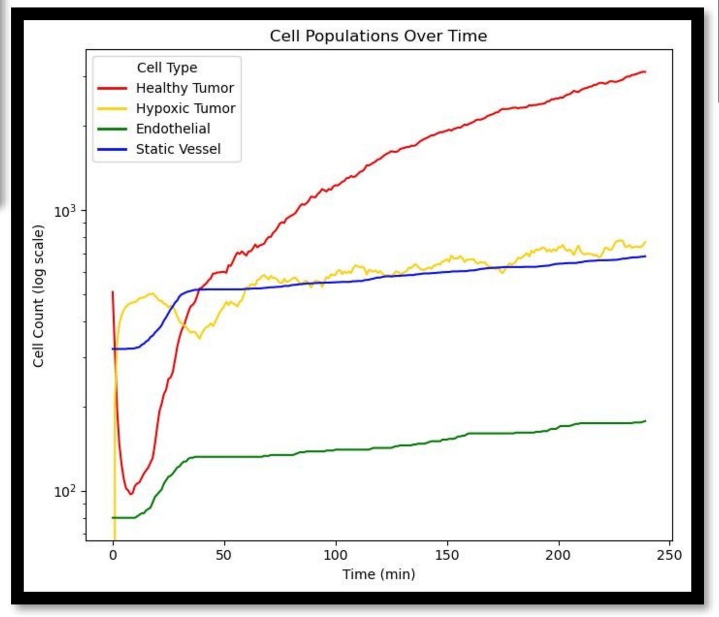
2 Days: Tumor secretes VEGF; blood vessels increase volume in response. Angiogenesis is triggered; vascular sprouts begin forming.



5 Days: Tumor cells reach blood vessels and access increased oxygen



10 Days: Tumor growth is now supported and sustained by vasculature



Cancer Cell Populations Plot: Tumor and vessel cell population dynamics over time

- Hypoxic Tumor
- Healthy Tumor
- Endothelial Cells
- Static Blood Vessels

Discussion and Future Work

Tumor has successfully manipulated its environment to grow vascular support

- By day 10, tumor expanded from the hypoxic center.
- Blood vessels deliver oxygen to nearby tumor cells.
- Oxygen gradient (heatmap) shows higher concentrations near vessels.
- Vascular support enables continued tumor growth.

Some limitations:

- Cancer travelled to existing vasculature – try to make vasculature travel to cancer too.
- Improve physical vessel–vessel and vessel–cancer interactions.

References

1. Gupta, M. K., & Qin, R.-Y. (2003, June). *Mechanism and its regulation of tumor-induced angiogenesis*. World journal of gastroenterology.
2. Jain RK, "Normalization of tumor vasculature: an emerging concept in antiangiogenic therapy" – Nature Medicine, 2005. Anderson, A. R. A., & Chaplain, M. A. J. (1998). Continuous and discrete mathematical models of tumor-induced angiogenesis. Bulletin of Mathematical Biology, 60(5), 857–899.
3. Rocha H.L. et al., "A multiscale model of immune surveillance in micrometastases" – Syst Biol Appl, 2024. Bauer, A. L., Jackson, T. L., & Jiang, Y. (2007). A cell-based model exhibiting branching and anastomosis during tumor-induced angiogenesis. Biophysical Journal, 92(9), 3105–3121.

Acknowledgements

We sincerely thank Mary Elizabeth Loveless for her invaluable guidance through this project. We also extend our gratitude to John Metzkar for his insightful lecture and demonstration on using PhysiCell, which greatly aided our simulation work.